

AgriSmart

Farm Intelligence & Management

Empowering small-scale farmers with data-driven insights.

The Challenge

- **Data Fragmentation:** Vital farm data (yields, costs, dates) is often kept in physical notebooks or not at all.
- **Market Access:** Difficulty in finding fair prices and reliable buyers/sellers.
- **Financial Visibility:** Poor tracking of ROI and seasonal cash flow.
- **Connectivity:** Rural areas often have unreliable internet access.

The Solution

AgriSmart is a **Local-First PWA** designed to give farmers total control over their operations, even in the heart of the field.

- **Offline Ready:** Works without internet via IndexedDB.
- **Comprehensive Tracking:** From seedling to sale.
- **Integrated Marketplace:** Direct farmer-to-buyer connections.

Key Features: Dashboard

- **Real-time KPIs:** Instant view of Active Crops, Revenue, and Profit.
- **Financial Trends:** Comparative charts of revenue vs. expenses.
- **Actionable Alerts:** Notifications for harvests and order updates.

Key Features: Crop Management

- **Lifecycle Tracking:** Monitor stages from *Seedling* to *Harvest*.
- **Yield Analytics:** Compare expected vs. actual yield to improve accuracy.
- **Input Tracking:** Detailed logs for seeds, fertilizers, and pesticides.

Key Features: Marketplace

- **Direct Listings:** Post produce availability with location and pricing.
- **Smart Orders:** Seamless order placement and status tracking.
- **Transparency:** Clear history of sales and purchases.

Technical Architecture

- **Frontend:** React + Vite (ESM).
- **Styling:** Vanilla CSS Modules (Modern & Lightweight).
- **Data Layer:** IndexedDB for local-first persistence.
- **PWA:** Service Workers for offline reliability and performance.
- **Auth & State:** React Context for robust state management.

Future Vision

- **IoT Integration:** Soil moisture and weather sensor connectivity.
- **AI Insights:** Predictive yield modeling based on historical data.
- **Micro-Financing:** Data-backed credit scoring for farm loans.

AgriSmart

Digitizing the Shamba.

Thank You!

Questions?